CoRoT-2b

R-0151

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_CoRoT-2_180608` · created 2026-07-16T09:47:56+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458277.8067 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.016
Transit depth (Rp/Rs)^2	2.28 +/- 0.49 [%]
Orbital Inclination (inc)	89.79 +/- 1.13
Airmass coefficient 1 (a1)	1.436 +/- 0.029
Airmass coefficient 2 (a2)	-0.1821 +/- 0.0046
Scatter in the residuals of the lightcurve fit is	2.0 %
Best Comparison Star	None
Optimal Aperture	4.75
Optimal Annulus	6.94
Transit Duration (day)	0.0951 +/- 0.0017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.016 vs 0.1667 (deviation 0.78σ)
Duration fit vs published	0.0951 d vs 0.095 d (deviation 0.05σ)
Mid-time O–C	4.6 min
Depth SNR	7.5
Residual scatter	2.0 %

Light curve

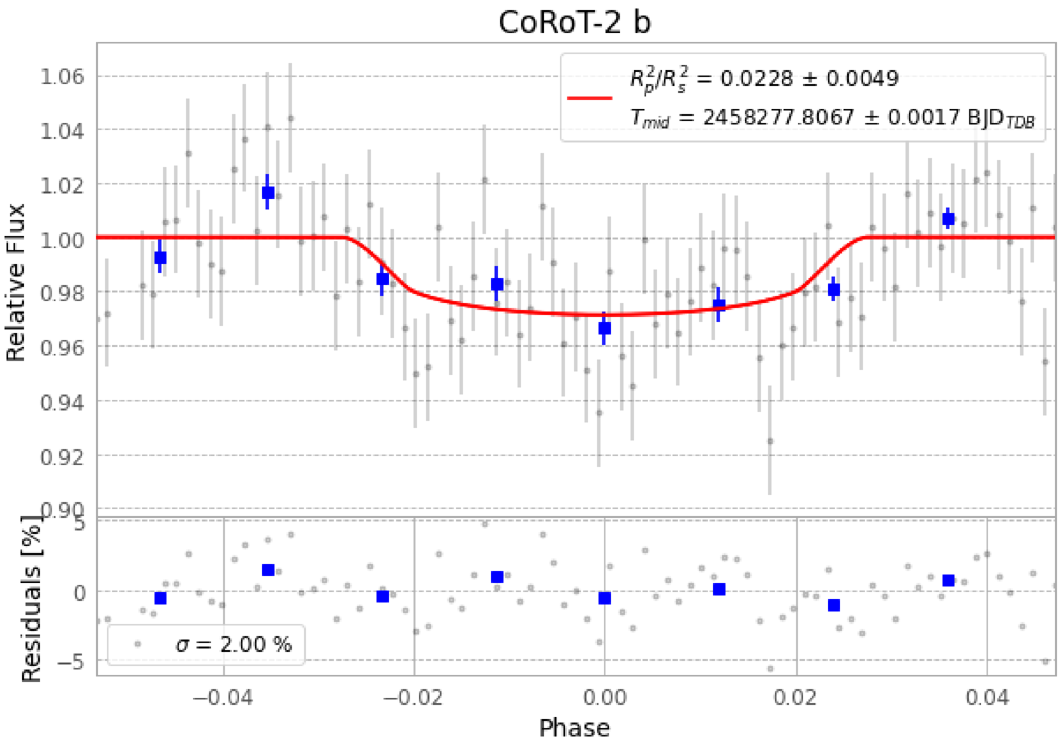

Figure 1 — FinalLightCurve_CoRoT-2 b_2018-06-07.png (SHA-256 c5f41049040a8df9…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [329, 153], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 98cbbdaa234a85be…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba`

Data provenance

86 FITS frames (56 MB), `CoRoT-2180608045712.FITS` → `CoRoT-2180608091212.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-180608.log (32eb7c8ca08a…), FinalLightCurve_CoRoT-2 b_2018-06-07.png (c5f41049040a…), FinalLightCurve_CoRoT-2 b_2018-06-07.csv (210c721c4a9f…)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-16 09:47Z.